

Design Framework for Sustainable Textile-Reinforced Concrete Using Laminate Theory and Damage Modeling

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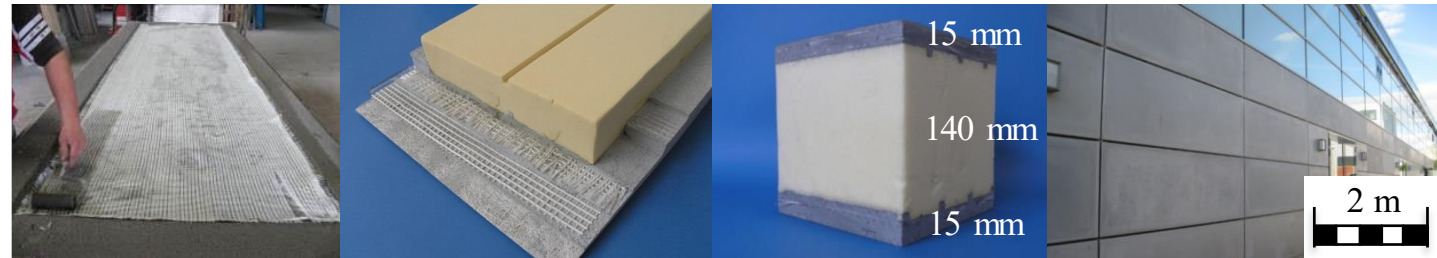
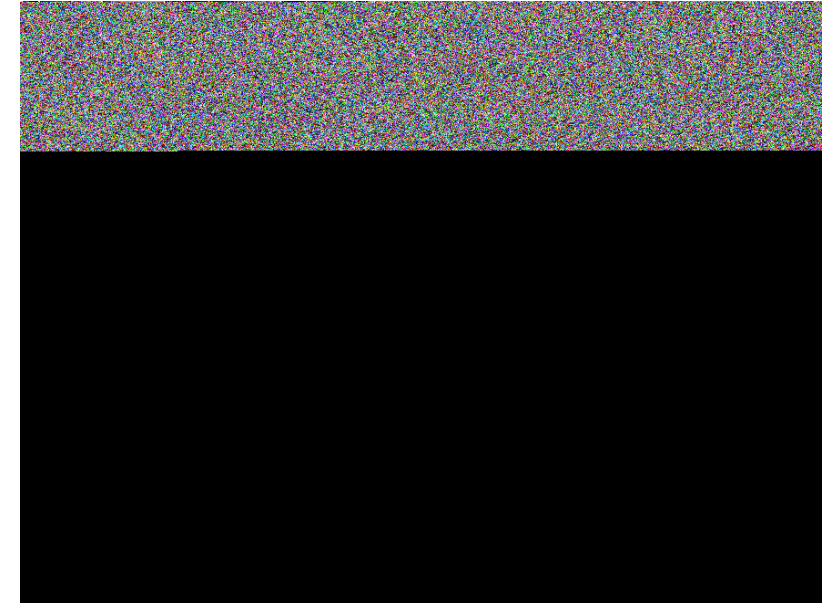
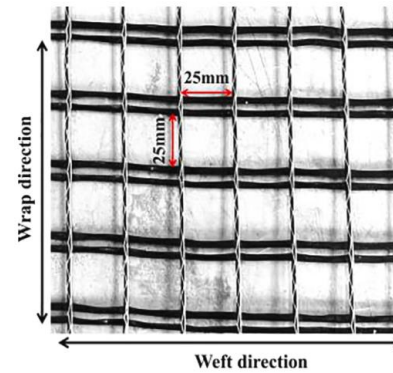
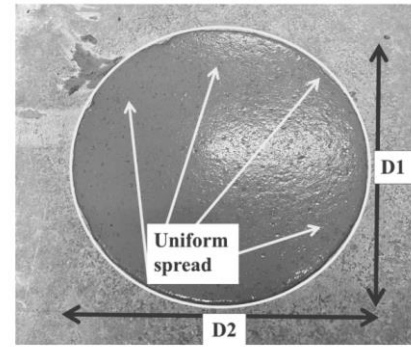
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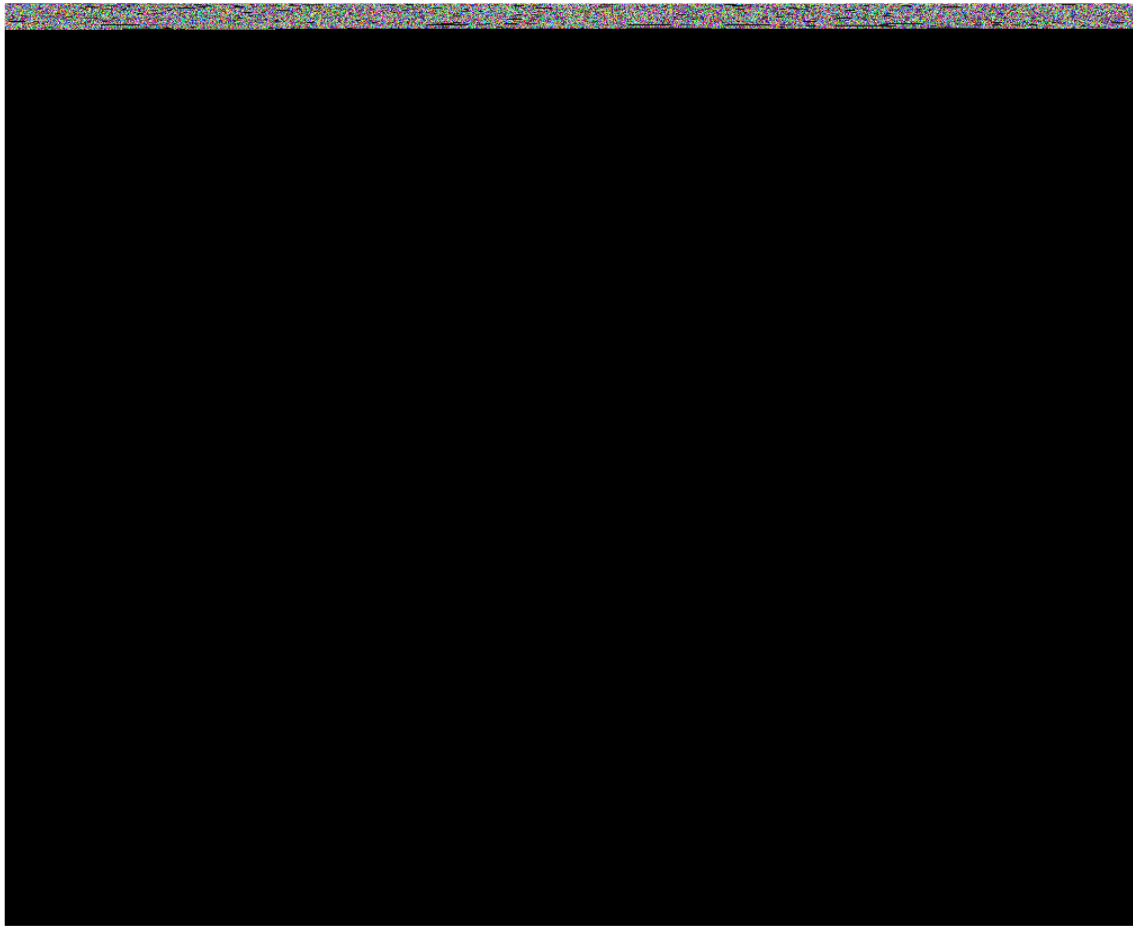


RILEM 2026 Spring Convention
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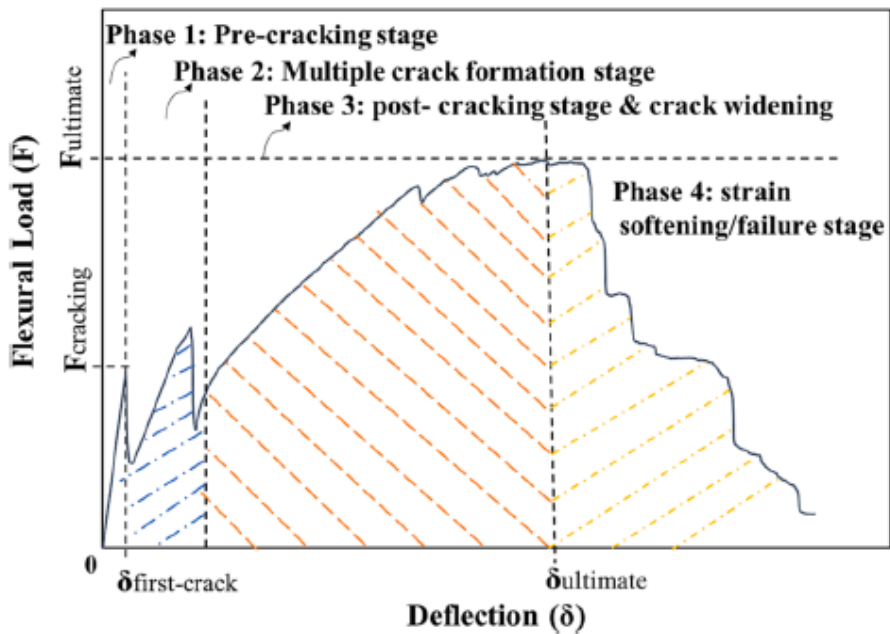
Basalt Textile-Reinforced Concrete

- High tensile-toughness characteristics
- Superior impact, earthquake, and fatigue characteristics.
- Prefabricated Structural elements, thin sheets, panels, cladding members.
- Structural repair and retrofit.
- Corrosion-resistant
- High temperature resistance vs. glass/carbon
- Lower cost than carbon fiber

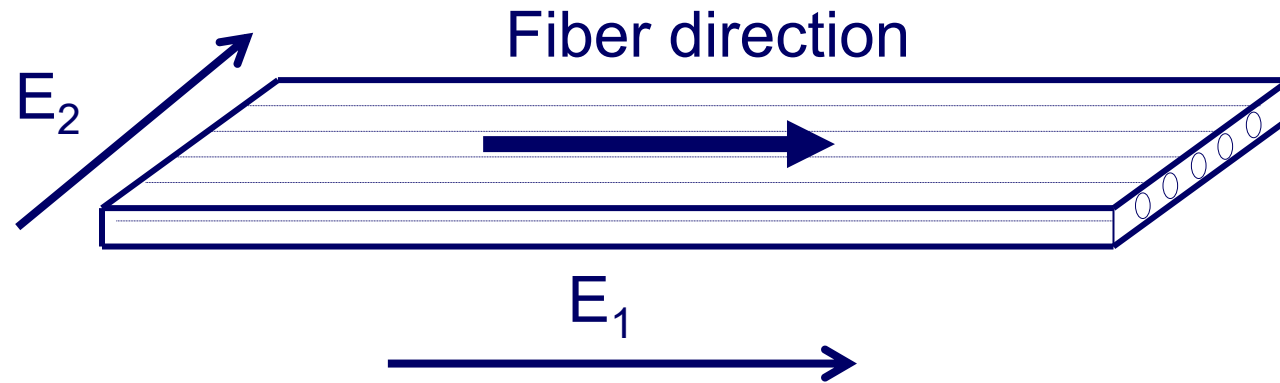




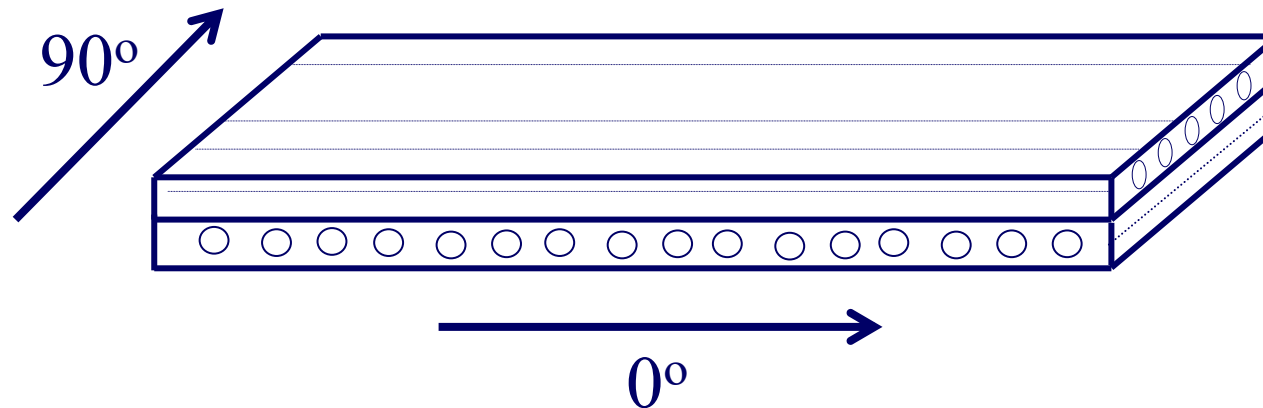
tensile reinforcement ratios (0.37%,
0.56% and 0.93%, C1, C6, C11



Sample ID	1L0P 0R28D C1	2L0P 0R28D C2	1L0P 0R28D C3	2L0P 0R7D C4B1	2L0P 0R28D C4B1	2L0P 0R7D C4B7	2L0P 0R28D C4B7	2L0P 0R28D C4B1S	2L0P 0R28D C5	2L0P 0R28D C6	3L0P0 R28D C7	2L0P 0R28D C8	3L0P 0R28D C9	3L0P 0R28D C10	1L0P 0R28D C11
Failure mode	F.R	F.R	P.R	C.C/ P.R	C.C/ P.R	C.C/ P.R	C.C/ P.R	F.R	C.C/ P.R	P.R	P.R	P.R	P.R	P.R	P.R
Non- prestressed plates	210,0	210,5	215,5	210P/R7D	215,10 210P/R28D	210P/R7D	210P/R28D	1R0P2LSC	215,15	210,5	310,2,5	310,5,5	310,5,10	310,5,15	510

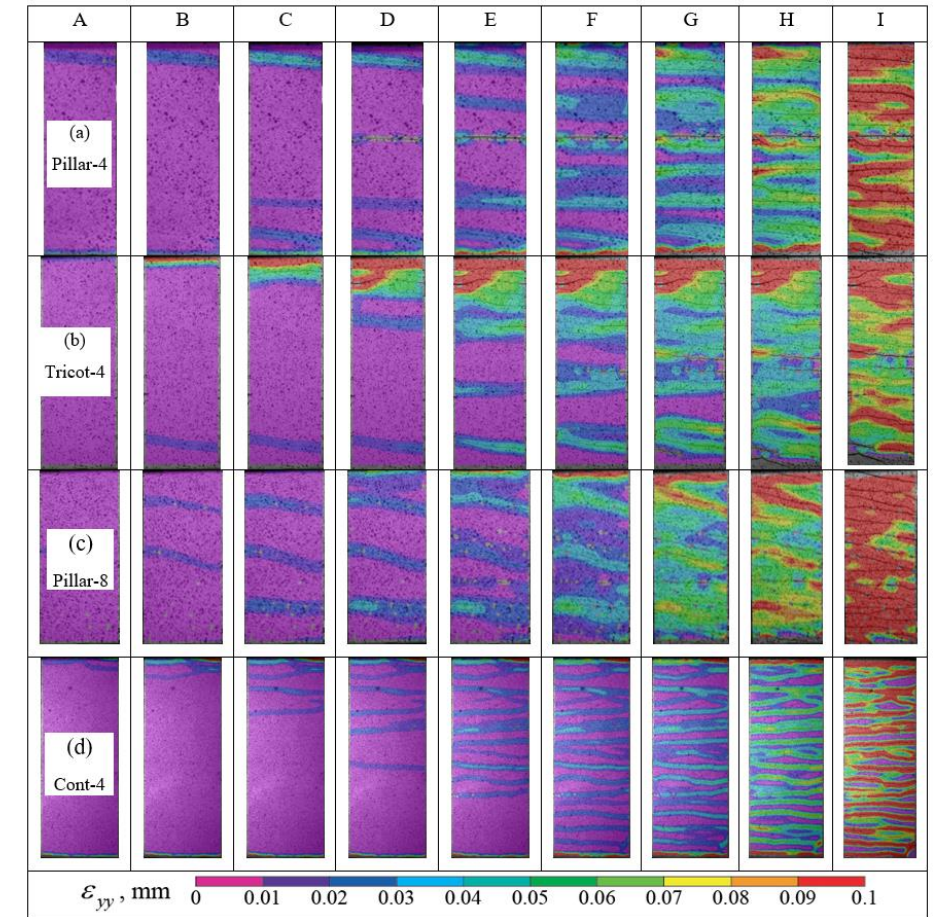
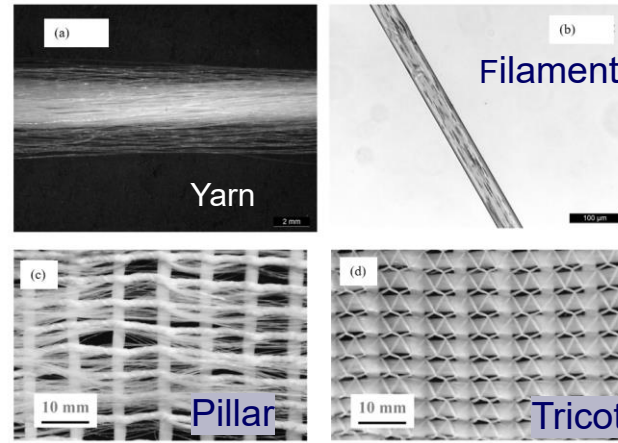
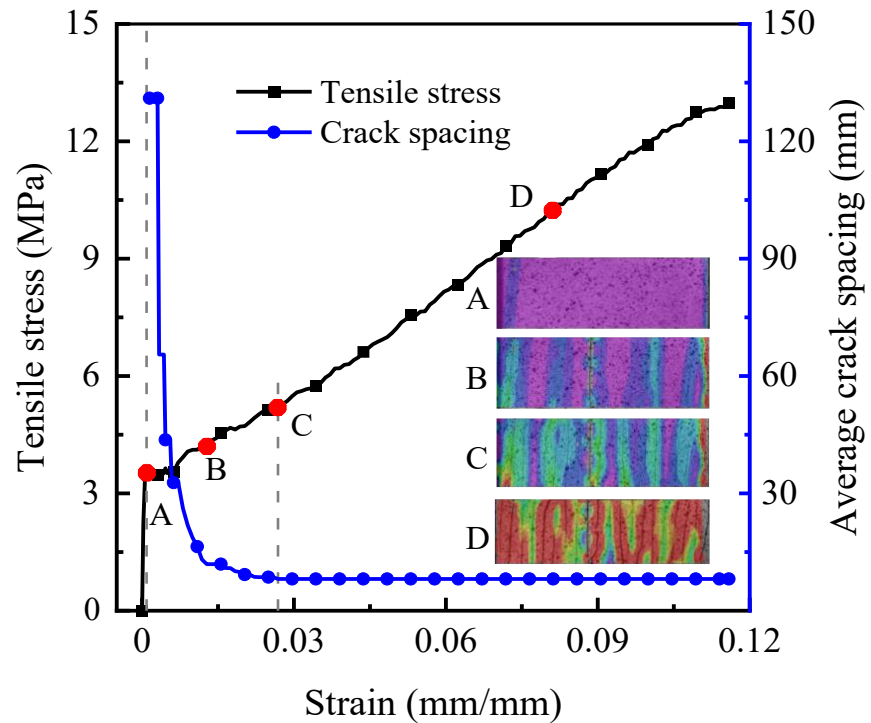


The unidirectional ply (or lamina) has maximum stiffness anisotropy - $E_1 \gg E_2$



We could remove the in-plane anisotropy by constructing a 'cross-ply' laminate, with UD plies oriented at 0 and 90° . Now $E_1 = E_2$

Distributed Cracking Mechanisms in TRC

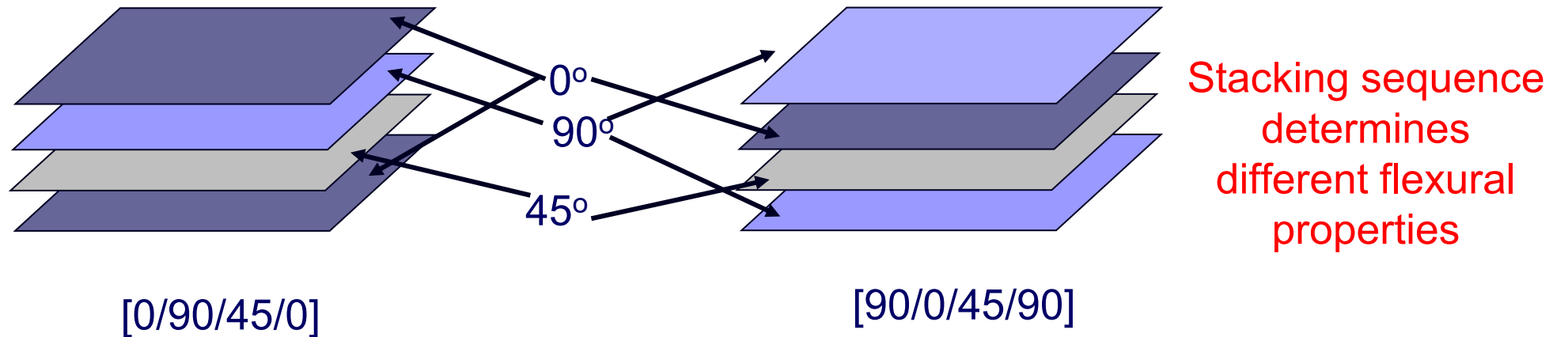


- DIC tracks each crack via time stamp, strain, and image intensity.
- New cracks form randomly (not sequentially).

Symmetric Laminates

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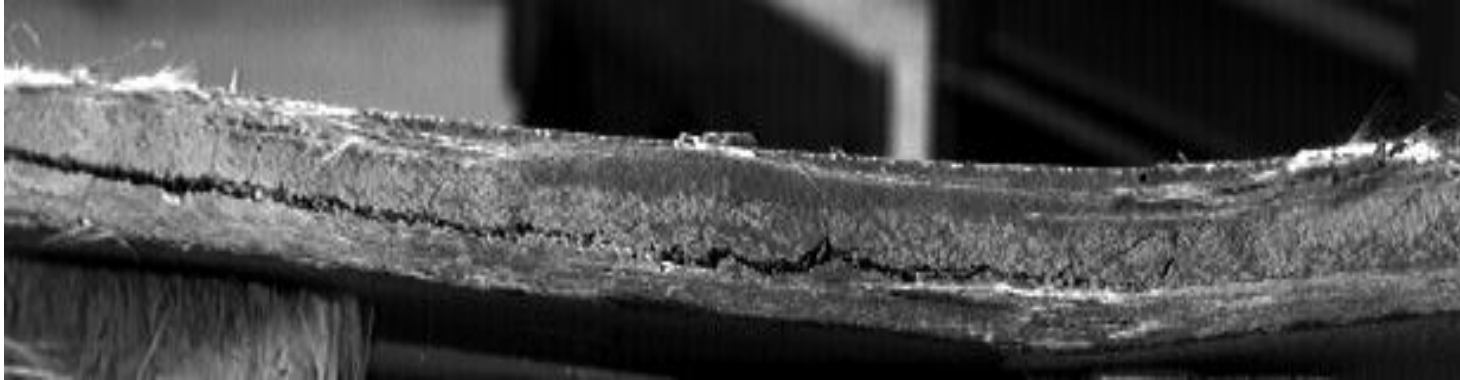
To avoid coupling effects, the cross-ply laminate must be *symmetric* - each ply must be mirrored (in terms of thickness and orientation) about the center.



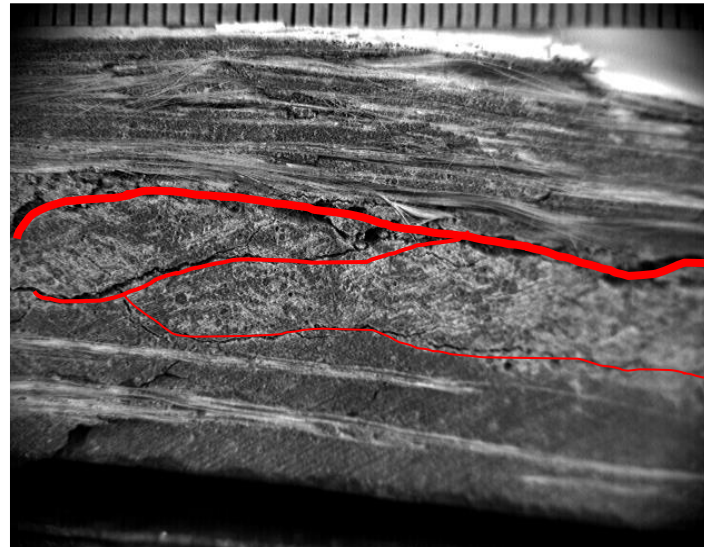
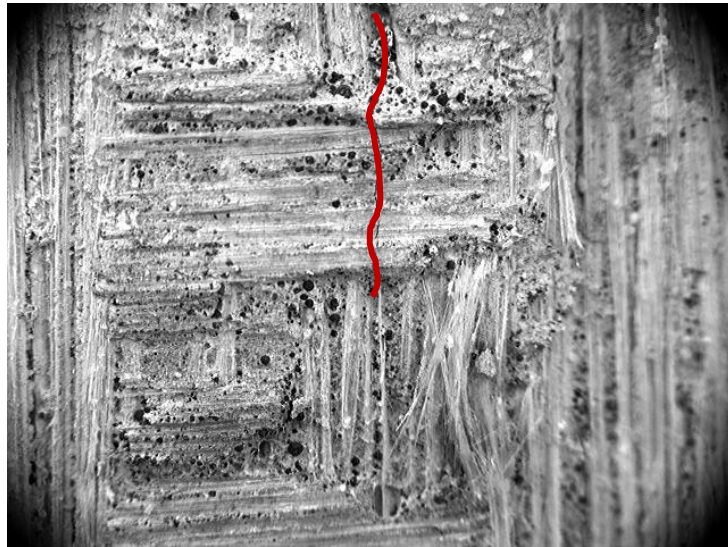
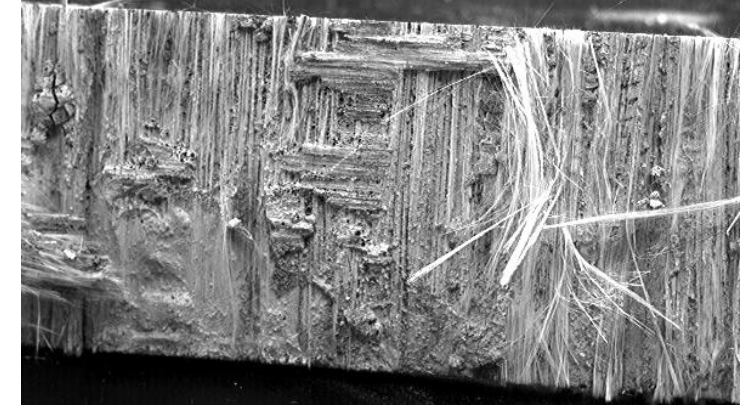
- Ply orientations determine the in-plane properties of the entire section.
- The two laminates $[0/90/45/0]$ and $[90/0/45/90]$ have the same in-plane stiffness, **but!** different flexural stiffnesses
- The $[0/90/45/0]$ laminate becomes $[90/0/45/90]$ if rotated. Therefore, this cross-ply laminate has flexural properties that depend on how the load is applied.

Weak Interfaces Result in Strong & Tough Composites

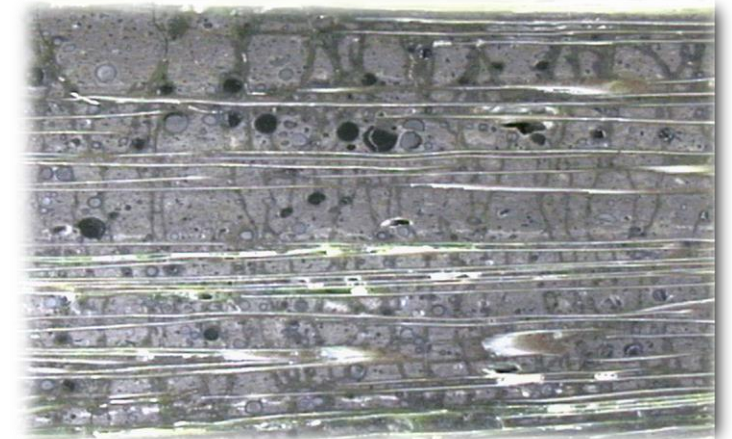
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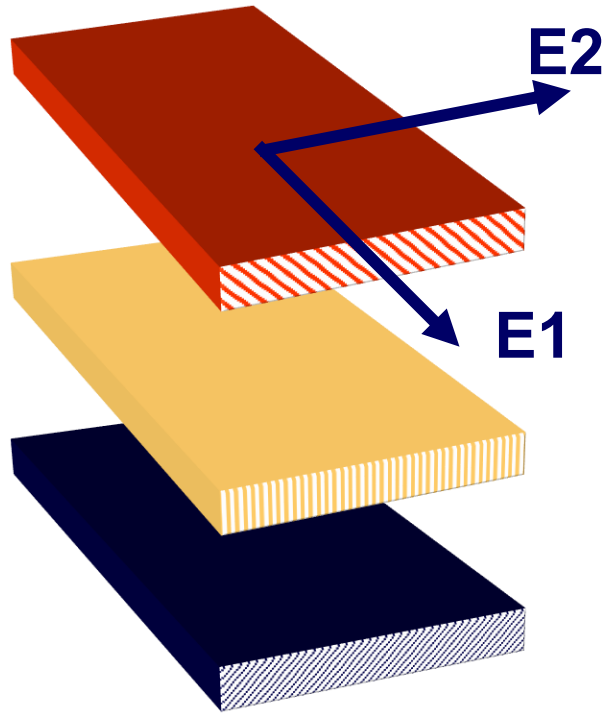
Cracked Laminate



Distributed microcracking in
unidirectional composite



Interfacial Microcracking in between plies is a toughening Mechanism



- Rule of Mixtures [Voigt model] for longitudinal stiffness

Composite ply matrix

↓ ↓ ↓

$$E_1(\omega) = \eta_f E_f V_f + E_m(\omega)(1 - V_f)$$

- Halpin-Tsai equations for transverse modulus E_2 , G_{12} , and ν_{12}

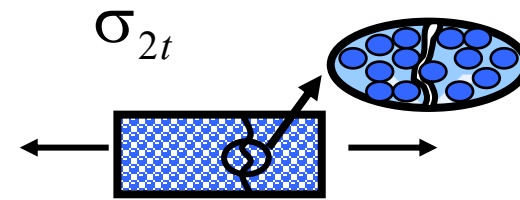
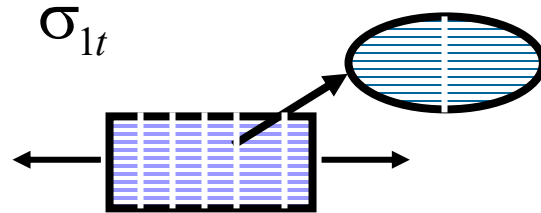
$$E_2(\omega) = \frac{E_m(\omega)(1 + \xi \eta V_f)}{1 - \eta V_f} \quad \eta = \frac{E_f - E_m(\omega)}{E_f + \xi E_m(\omega)}$$

ω is the damage parameter, $0 \leq \omega_i \leq 1$

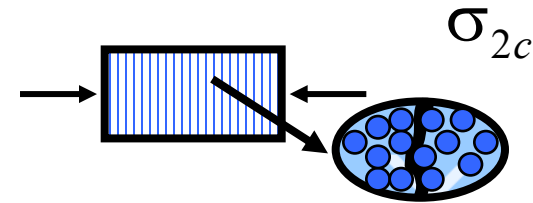
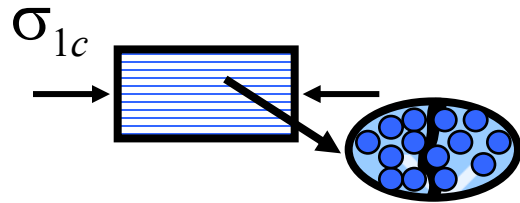
Modes of Failure

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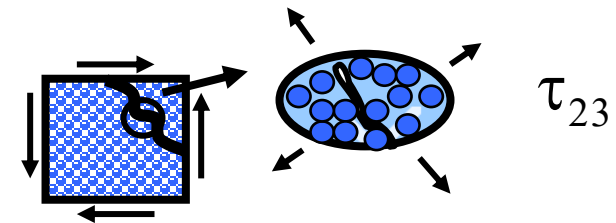
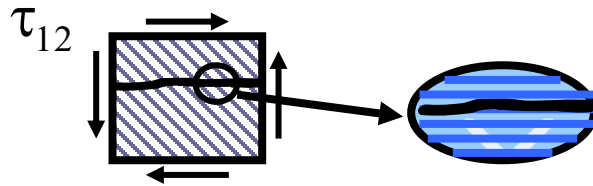
Tension



Compression



Shear



Failure Criteria and Damage Parameter

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Tsai-Wu failure criterion

$$F(\sigma_1, \sigma_2, \tau_{12}) = F_{11}\sigma_1^2 + 2F_{11}\sigma_1\sigma_2 + F_{22}\sigma_2^2 + F_{66}\sigma_{12}^2 + F_1\sigma_1 + F_2\sigma_2 = 1$$

$$F_1 = \frac{1}{|\sigma_{1t}|} - \frac{1}{|\sigma_{1c}|} \quad F_2 = \frac{1}{|\sigma_{2t}|} - \frac{1}{|\sigma_{2c}|} \quad F_{11} = \frac{1}{|\sigma_{1t}\sigma_{1c}|} \quad F_{22} = \frac{1}{|\sigma_{2t}\sigma_{2c}|} \quad F_{66} = \frac{1}{\sigma_{6u}^2}$$

Tsai-Wu failure criterion is calculated to verify whether matrix failure has initiated.

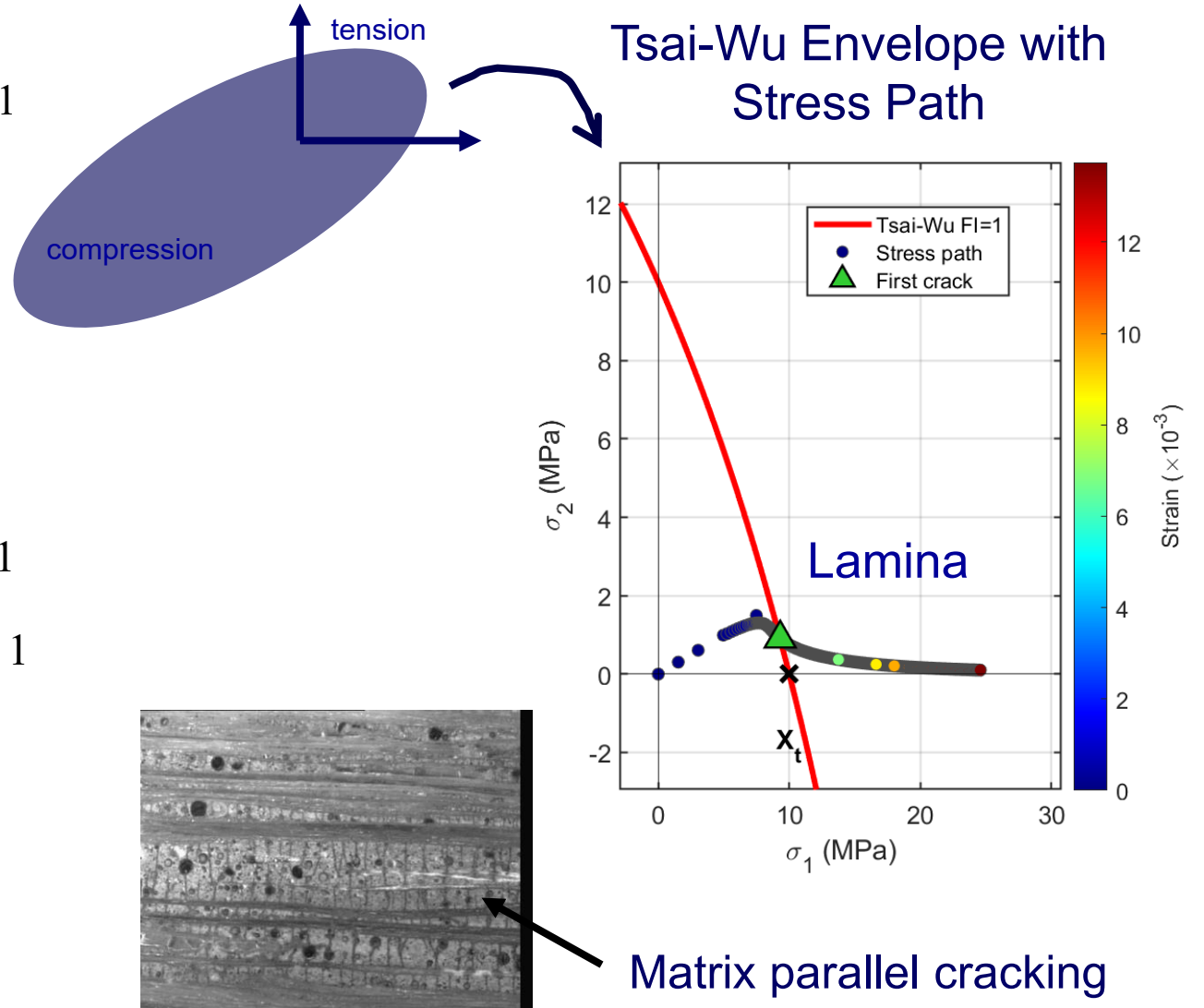
$$\omega_i^k = \begin{cases} 0 & \forall F(\sigma_1, \sigma_2, \tau_{12}) \leq 1 \\ 1 - e^{-\alpha(\varepsilon_1 - \varepsilon_{cr})^\beta}, & 0 \leq \omega_i \leq 1 \quad \forall F(\sigma_1, \sigma_2, \tau_{12}) > 1 \end{cases}$$

Degradation of E matrix

$$\bar{E}_m(\omega) = \frac{\bar{E}_m}{1 + \frac{16}{3}\omega(1 - \nu_m^2)}$$

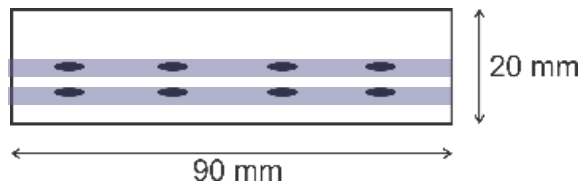
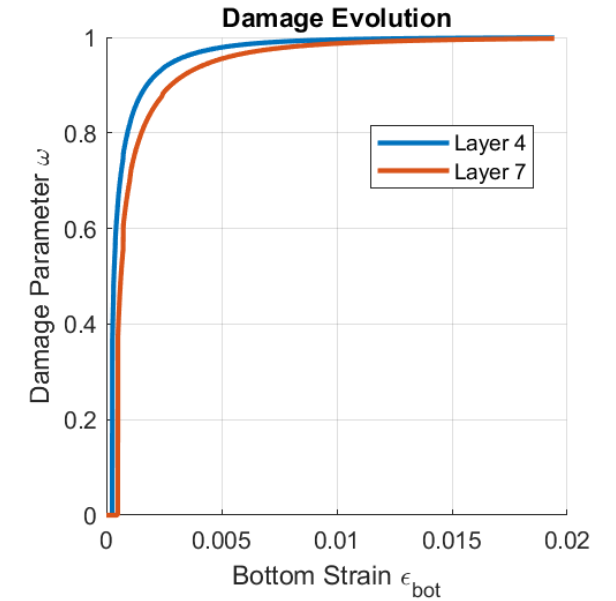
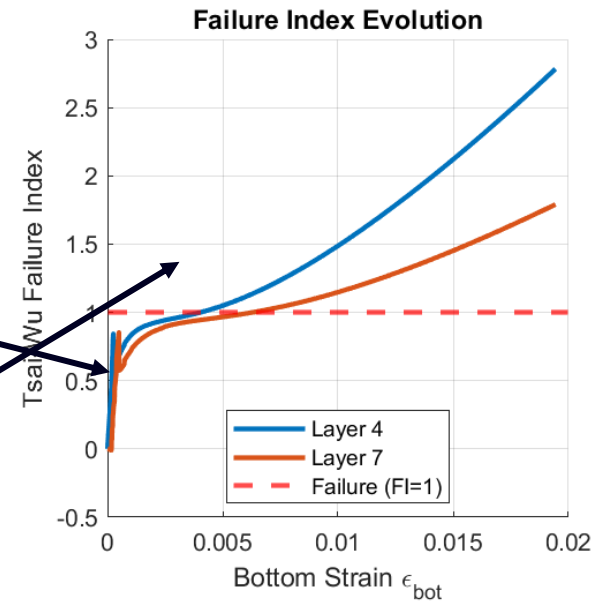
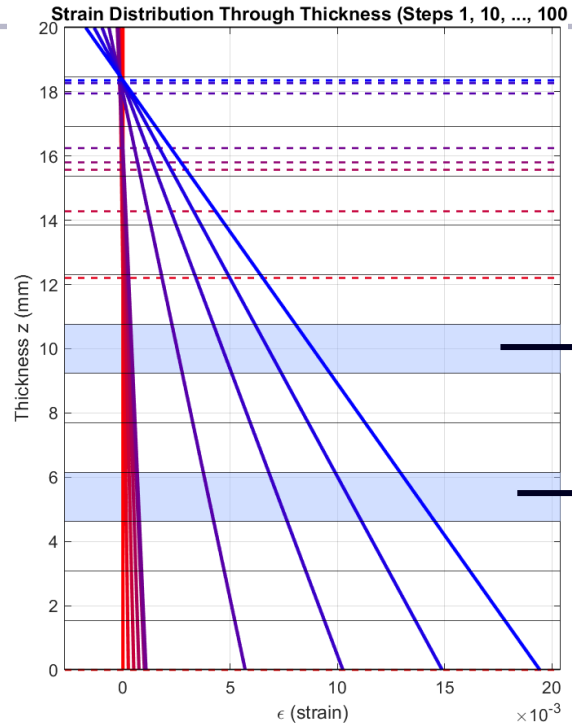
or

$$\bar{E}_m(\omega) = \bar{E}_m(1 - \omega)$$

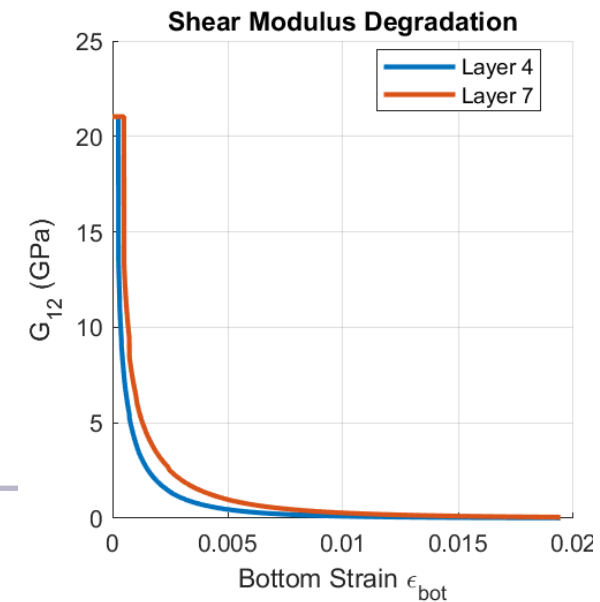
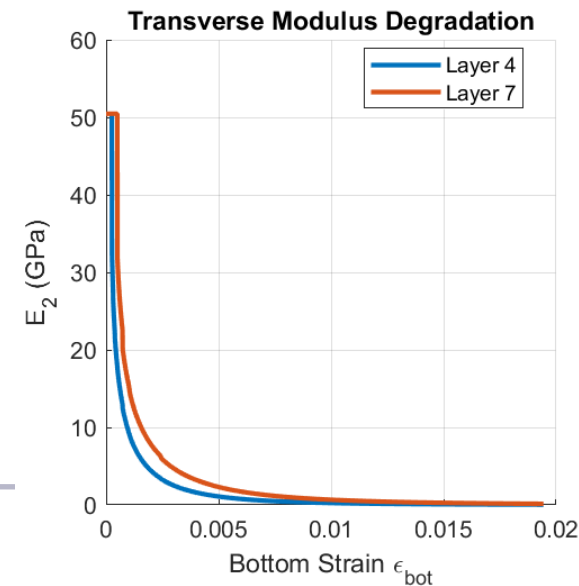
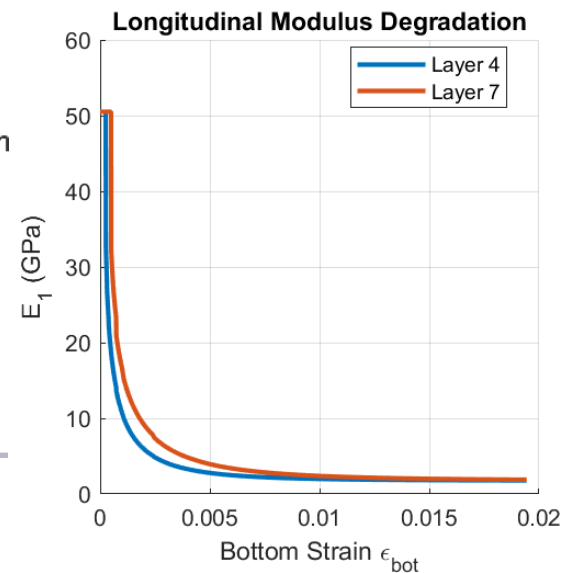


Progressive Damage Analysis

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TRC cross-section



Orientation Effects and Stress Transformation

The stiffness along an arbitrary orientation θ is denoted as $\bar{Q}_{ij}$

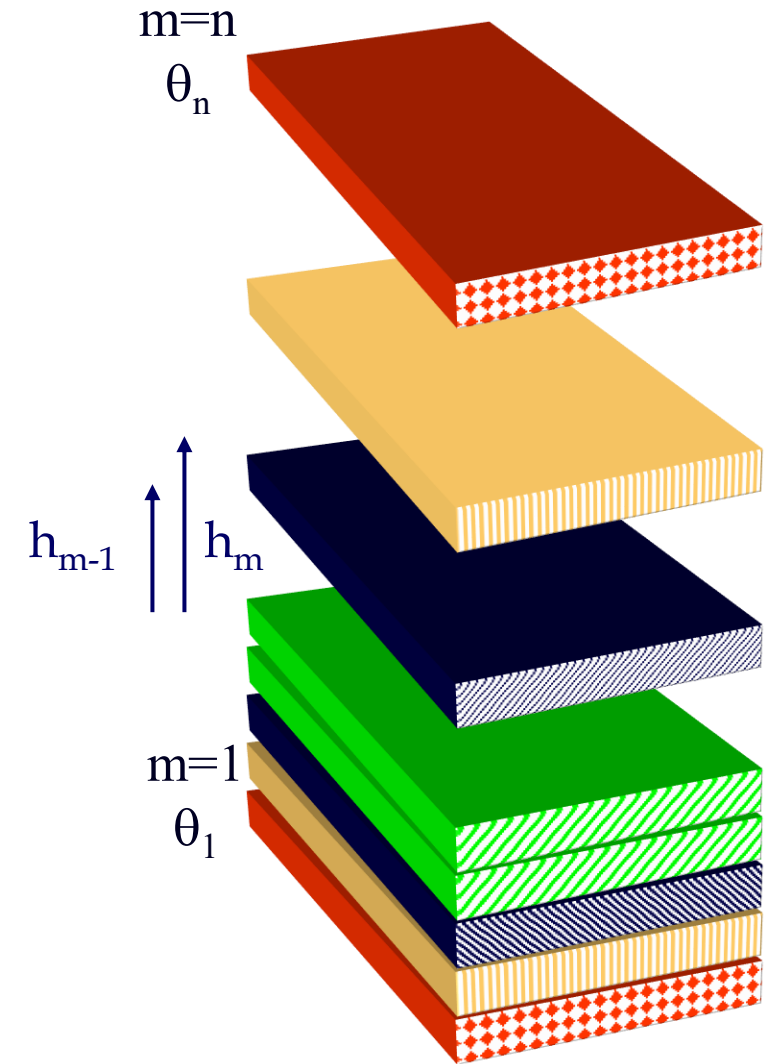
$$\bar{Q}_{ij} = T_{ij} R Q_{ij} R T_{ij}^{-1} \quad \text{where}$$

Reuter Matrix

coordinate transformation, T

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = T_{ij} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} \cos^2\theta & \sin^2\theta & 2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & -2\sin\theta\cos\theta \\ -\sin\theta\cos\theta & \sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}$$

For a composite laminate consisting of several lamina each with an orientation of θ_m , where m represents the first to the n ply



Incremental Approach to Laminate Formulation

Laminate Stiffness

Uniaxial loads Bending

$$\begin{bmatrix} \Delta N \\ \Delta M \end{bmatrix} = \begin{bmatrix} \bar{A} & \bar{B} \\ \bar{B} & \bar{D} \end{bmatrix} \begin{bmatrix} \Delta \varepsilon^0 \\ \Delta \kappa \end{bmatrix}$$

← Changing in Uniaxial strain

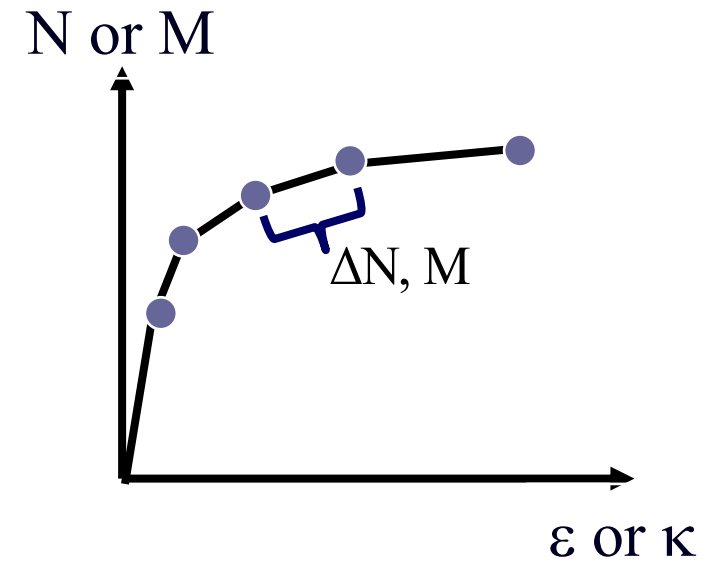
← Changing in Curvature

The classical lamination theory results in derivation of lamina stiffness components as

$$A_{ij} = \sum_{m=1}^n \bar{Q}_{ij}^m (h_m - h_{m-1})$$

$$B_{ij} = \frac{1}{2} \sum_{m=1}^n \bar{Q}_{ij}^m (h_m^2 - h_{m-1}^2)$$

$$D_{ij} = \frac{1}{3} \sum_{m=1}^n \bar{Q}_{ij}^m (h_m^3 - h_{m-1}^3)$$



For uniaxial loads in a symmetric lamina (B=0):

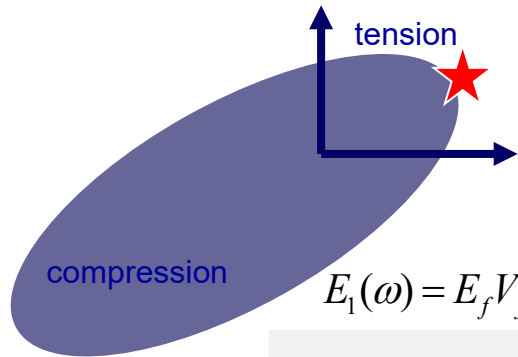
$$\begin{bmatrix} \Delta N \end{bmatrix} = \begin{bmatrix} \bar{A} \end{bmatrix} \begin{bmatrix} \Delta \varepsilon^0 \end{bmatrix}$$

For bending only in a symmetric lamina (B=0):

$$\begin{bmatrix} \Delta M \end{bmatrix} = \begin{bmatrix} \bar{D} \end{bmatrix} \begin{bmatrix} \Delta \kappa \end{bmatrix}$$

Computations Within a load step

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$$E_1(\omega) = E_f V_f + E_m(\omega)(1 - V_f)$$

$$\bar{Q}_{ij} = T_{ij}^R Q_{ij}^R T_{ij}^{-1}$$

$$A_{ij} = \sum_{m=1}^n \bar{Q}_{ij}^m (h_m - h_{m-1})$$

$$B_{ij} = \frac{1}{2} \sum_{m=1}^n \bar{Q}_{ij}^m (h_m^2 - h_{m-1}^2)$$

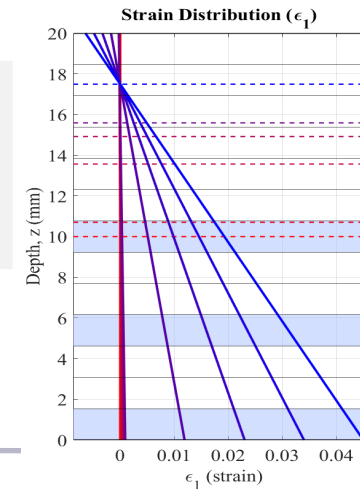
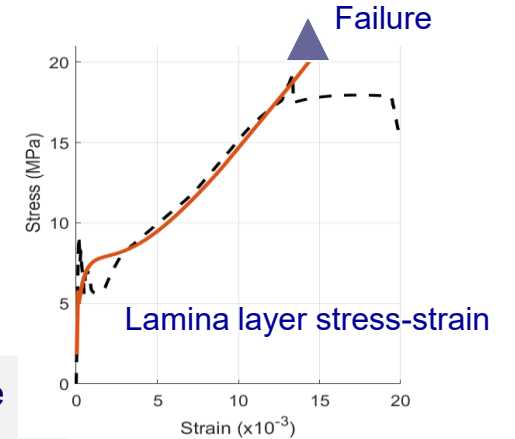
$$D_{ij} = \frac{1}{3} \sum_{m=1}^n \bar{Q}_{ij}^m (h_m^3 - h_{m-1}^3)$$

Compute overall stiffness
Calculate the strains, stresses,
curvature, and elongation

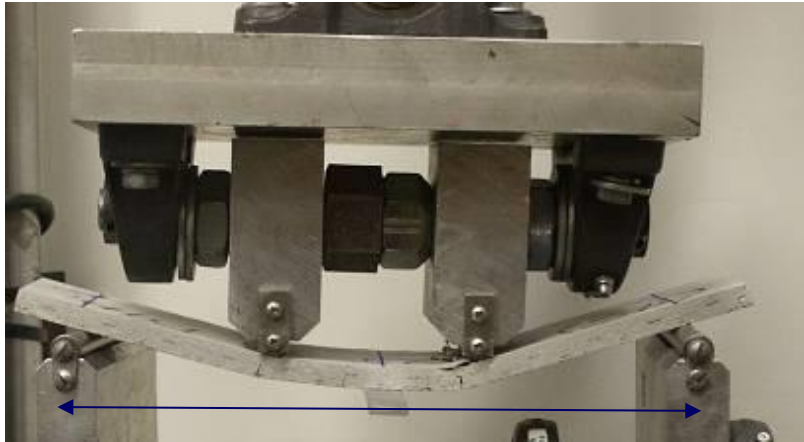
Check for the failure in the components
of the lamina using the failure criteria
Using Tsai-Wu
update the lamina properties

Check for the failure
of the entire lamina
using ultimate strength
update the structure

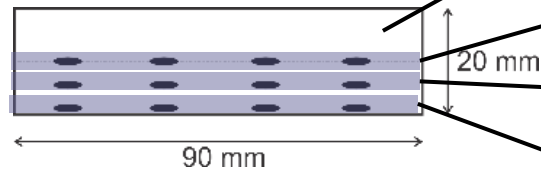
Initialize all the geometrical
and historical parameters for each
lamina. Impose the strain distribution,
update the stresses in this increment



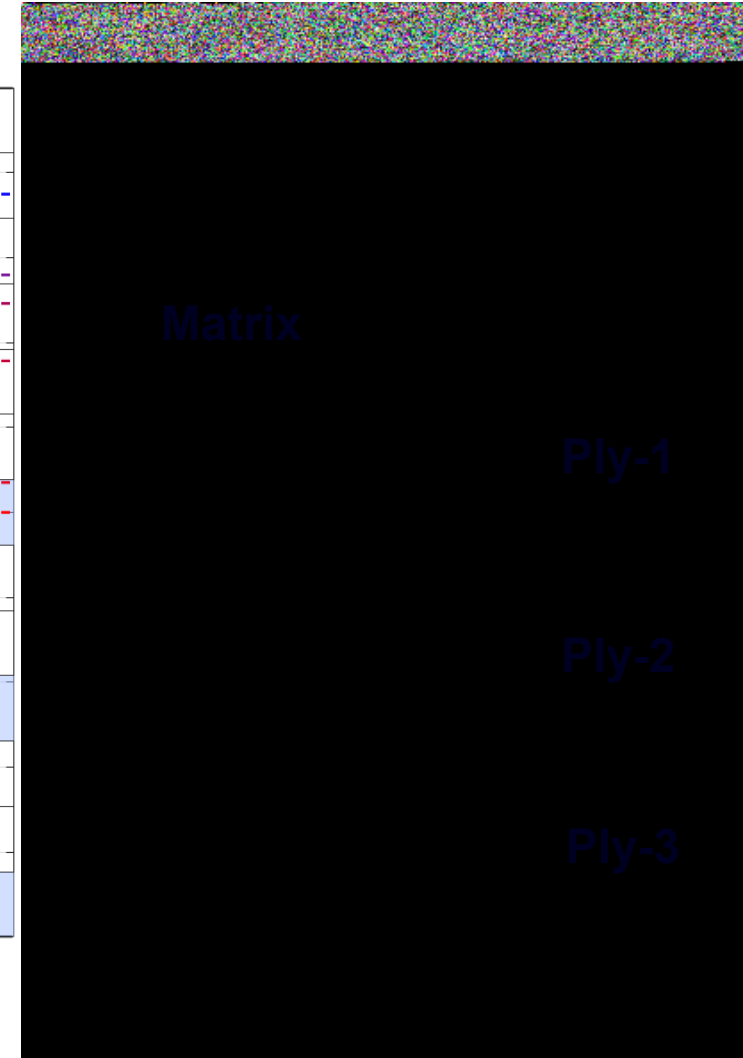
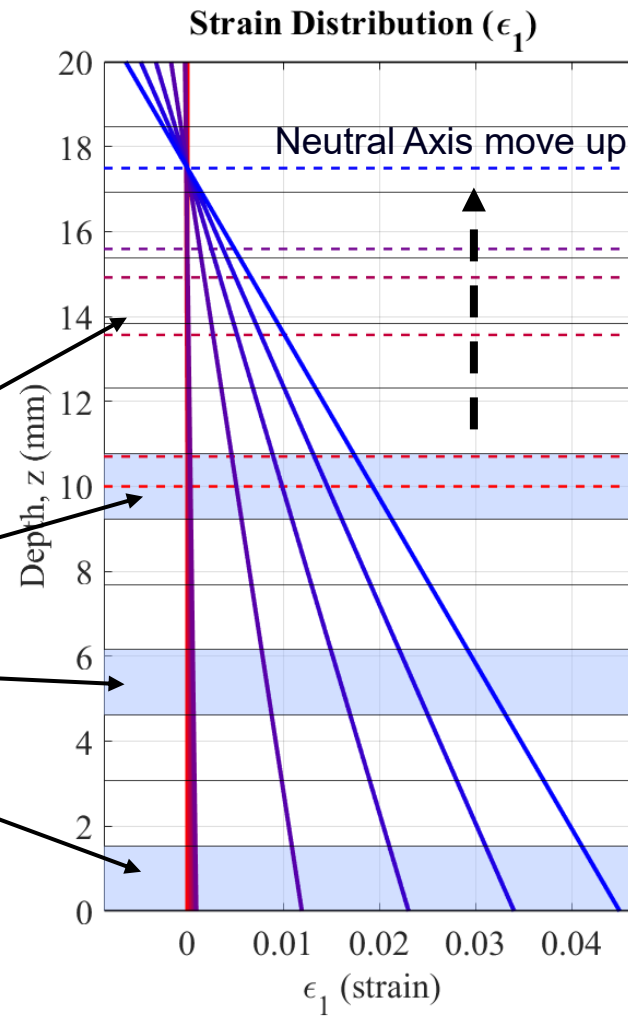
Strain and Stress Distribution in Flexural Sample



Flexural test

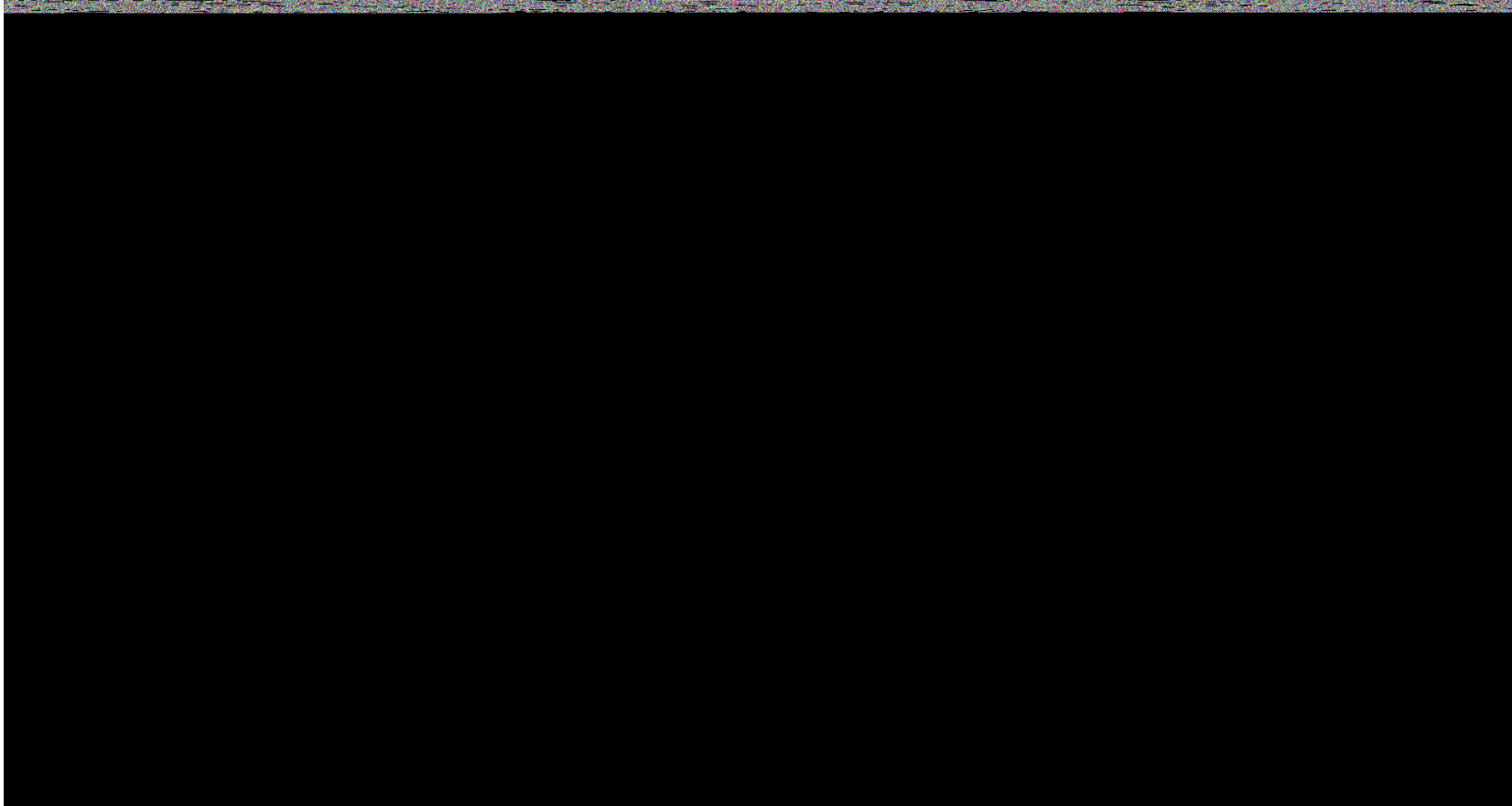


TRC cross-section



Comparison of load–deflection responses for BTRC

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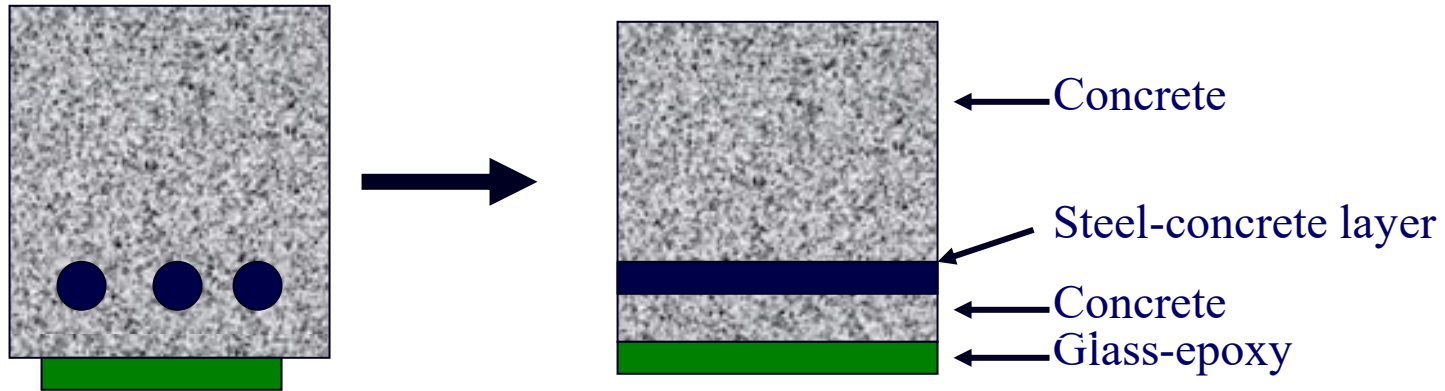


Results of Individual Series

- Comparison of simulated and experimental results

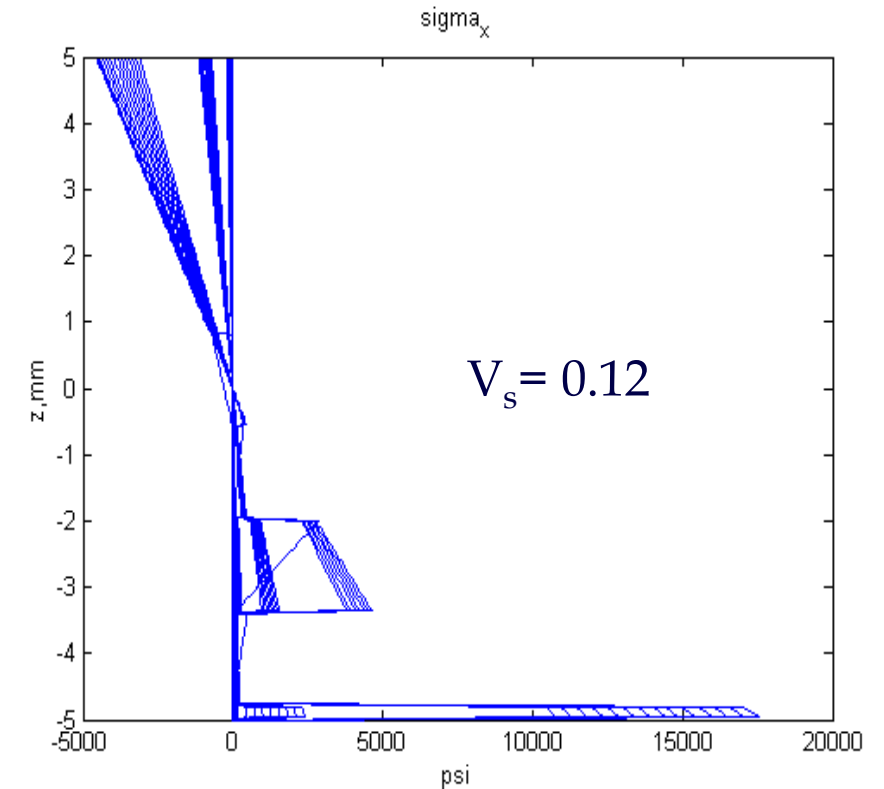
Composite Retrofit - Reinforced Concrete beam-Glass Epoxy Bonded Laminate

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material	Young's Modulus psi	Posisson's ratio	Compressive Strength, psi	Tensile strength, Psi	Coefficient of Thermal expansion
epoxy	6.e5	0.28	20000	6000	2.6e-6
glass fibers	11.5e6	0.25	200000	200000	1.e-6
paste	3.e6	0.17	8000.	500.	1.e-6
steel	29.e6	0.3	36000	36000	6.e-6

Stress Distribution in Flexural Sample



- Theoretical models based on composite laminate Theory can be used to predict the mechanical response of various cement based composite systems.
- Combination of damage mechanics based methods and composite laminate theory can result in a useful method to evaluate the response of various fiber matrix, geometry, and loading combinations.
- The formulation is applicable to a wide range of materials studied including unidirectional, angle ply, sandwich, fabric reinforced, and also retrofit composite systems.